BSc in Petroleum Geosciences and Remote Sensing

Department of Applied Physics and Astronomy

College of Sciences

Personnel

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Vision

The program aims to be recognized in the region in providing highly qualified graduates who can address the rapid technological challenges of the future.

Mission

The program is committed to graduate highly qualified professionals equipped with latest knowledge in Petroleum, Geosciences and Remote Sensing areas and skills who can contribute to the economic development of the United Arab Emirates and the region.

Program Goals

1. Provide students with the knowledge and skills, including problem analysis, solving, and design, necessary for a successful career in Petroleum Geosciences and remote sensing.
2. Equip students with skills of critical thinking, teamwork, leadership and communications, and use them to solve complex problems in Petroleum Geosciences and remote sensing.
3. Prepare students to develop knowledge using modern design tools and new technologies in sciences and learn through appropriate lifelong education processes.
4. Prepare students to be admitted to and succeed in graduate study in internationally recognized universities.

Program outcomes

Upon successful completion of the B.Sc. in Petroleum Geosciences and Remote Sensing program the student will have the ability to:

a. Identify, formulate, and solve petroleum, geosciences and remote sensing problems.
b. Apply fundamental principles and concepts of geosciences and remote sensing in theoretical and practical situations.
c. Employ modern technologies and established IT skills to collect, interpret, and present geological data.
d. Implement independent experiments under guidance using to the appropriate research methodologies.
e. Work effectively, responsibly and safely in an individual or team context.
f. Communicate information concisely and accurately using written, visual, and verbal means appropriate to the situation.
g. Actively engaging in professional development and life-long learning activities.
h. Model geosciences related components to meet economic, environmental, social, political, ethical, health and safety needs.

Career Opportunities

Graduates from the PGRS will find employment opportunities covering a wide spectrum mainly in petroleum sectors. Geoscientists are in high demand in many oil and gas industries, mineral and water exploration, geo- imaging and remote sensing, natural risks management, environmental monitoring, forensic geosciences and archaeological excavation and preservation, as well as research positions in industry and government laboratories. Problem-solving techniques learned in the petroleum geosciences curriculum create opportunities for continued educational pursuits and/or higher graduate study in geosciences. The main potential employers and probable outlets for Geoscientists in the UAE are:

1. National and international oil and gas companies such as ADNOC, ADMA and ADCO, ENOC, Dubai Oil and gas, ExxonMobil, Shell, BP, Total and Schlumberger.
2. Governmental agencies with activities related to soil, water, energy, and environment; such as Ministry of Environment and Water, Ministry of Energy, Abu Dhabi and other Environmental Agency as well as Municipalities in all the Emirates.
3. Academic and educational institutions.
4. Civil Engineering Bureaus, construction companies and cement and rock quarries industry at both large and small scale entrepreneurs.
5. Police authority in forensic laboratories.
6. Archaeological and Heritage Museums.

Program Overview

The program is designed in accordance with the mission and vision of the University and the College of Science to meet the needs of the students, the community and the industry at large. The contents of the program are in line with or similar to many universities' undergraduate programs in USA and in Europe. Quality and excellence in both the curriculum and instructional pedagogy are ensured by following the procedures listed in table below. A student undertaking this program should complete a total of 123 credit hours distributed as follows:

	Compulsory	Support	Elective		Total
University Requirements	15	-	9		24
College Requirements	15	-	-		15
Department Requirements	50	13	**Dep.** 15	**Free** 6	84
Total	80	13	30		123

Course Numbering System: Each course number consists of 7 digits that are grouped in 5 fields as follows:

College No.		Dept. No.		Course level	Specialized field	Serial No.
1	4	6	0	X	Y	Z

Specialized fields of knowledge in Petroleum Geosciences and remote sensing are numbered as follows:

Digit	Field of Knowledge
1	Petroleum
2	Geology
3	Remote Sensing
0	General [seminar ,training , project]

I. University Requirements

The list of the University required courses and their descriptions are presented in the introductory pages of the College of Sciences section in this bulletin.

II. College Requirements

The list of the College required courses and their descriptions are presented in the introductory pages of the College of Sciences section in this bulletin.

III. Program Requirements

The program requirements consist of 84 credit hours of courses divided into four major sets as described below.

A. Mandatory Core Courses

This set consists of 50 credit hours encompassing the courses listed below:

Course #	Course Name	CrHrs	Prerequisites
1460120	Physical Geology	3	None
1460220	Sedimentary Rocks and Sedimentology	3	1460120
1460221	Stratigraphy and Structural Geology	3	1460120
1460223	General Geophysics	3	1460120, 1430117
1460224	Regional Geology	3	1460220, 1460221

1460230	Introduction to Geospatial Information System (GIS)	3	None
1460231	Introduction to Remote Sensing	3	1430117
1460310	Petrophysics	3	1460220
1460311	Petroleum Geology	3	1460220, 1460221
1460312	Exploration Geophysics I	3	1460223
1460313	Exploration Geophysics II	3	1460223
1460320	Geologic Remote Sensing	3	1460231
1460322	Seismology and Plate Tectonics	3	1460223, 1440241
1460400	Training	1	146311 & 65 credit hours completed
1460401	Seminar	1	Senior Standing
1460402	Graduation Project	3	Department consent
1460413	Reservoir Characterization	3	1460420
1460420	Well Logging	3	1460310

B. Supportive Core Courses

This category includes 13 credit hours offered by other departments for the students as indicated in the table below:

Course #	Course Name	CrHrs	Prerequisites
0202207	Technical Writing	3	None
1440132	Calculus II	3	1440131
1430117	Physics II	3	1440131(P/Co), 1430110
1430118	Physics II Lab	1	1430116, 1430117(P/Co)
1440241	Ordinary Differential Equation I	3	1440132

C. Elective Courses

These elective courses (15 credit hours) are to be chosen from a list of courses offered by the department. The support and core courses are the preparatory courses, which are designed to meet the breadth requirement in the program. After completing the preparatory courses, students are strongly encouraged to choose from alternative groupings of electives (referred to as “groups”) in different areas of PGRS to fulfill the depth requirement.

Course #	Course Name	CrHrs	Prerequisites
1460222	Paleontology	3	1460220
1460225	Igneous and Metamorphic Petrology	3	1460120
1460226	Environmental Geology and Risk Management	3	1460120
1460300	Special Topics I	3	Departmental approval
1460301	Special Topics II	3	Departmental approval
1460314	Petroleum Geology of the Middle East	3	1460220
1460324	Hydrogeology	3	1460120, 1460220, 1440131
1460326	Advanced Geographic Information System	3	1460230, 1411116
1460414	Petroleum Geochemistry	3	1460311
1460415	Basin Analysis	3	1460420, 1460221
1460421	Seismic Stratigraphy	3	1460221 ,1460312
1460431	Environmental Remote Sensing	3	1460231
1460432	Digital Image Processing	3	1440132

D. General Free Elective Courses

Such courses can be taken from the university’s pool of courses at large upon the approval of the academic advisor. These courses are intended to broaden the knowledge of students by combining studies from the program with studies from other academic disciplines.

Study Plan

The Petroleum Geosciences and Remote Sensing program encompasses 123 credits hours that are spread over eight semesters and could be completed in four years. The following distribution of courses by semester facilitated student's normal progression through the study plan.

Year 1, Semester 1 (16 Credits)			
Course	**Title**	**CrHrs**	**Prerequisites**
0201102 0201105	Arabic Language or Arabic for non-Arabic speakers	3	None
0202112	English for Academic Purposes	3	None
0202207	Technical Writing	3	None
1430110	Physics I for Sciences	3	None
1430116	Physics I Lab	1	1430110 (P/Co)
1440131	Calculus I	3	None

Year 1, Semester 2 (17 Credits)			
Course	**Title**	**CrHrs**	**Prerequisites**
1501110	AI and Digital Technologies	3	None
1420101	General Chemistry I	3	None
1420102	General Chemistry I Lab	1	1420101 (P/Co)
1430117	Physics II	3	1430117 (P/Co)
1430118	Physics II Lab	1	1430118(P/Co), 1430211(P/Co)
1440132	Calculus II	3	1440131
1460120	Physical Geology	3	None

Year 2, Semester 3 (16 Credits)			
Course	**Title**	**CrHrs**	**Prerequisites**
1411116	Programming I	4	None
1460220	Sedimentary Rocks and Sedimentology	3	1460120
1460221	Stratigraphy and Structural Geology	3	1460120
0104101	Islamic Culture	3	None
1460230	Introduction to Geospatial Information System (GIS)	3	None

Year 2, Semester 4 (15 Credits)			
Course	**Title**	**CrHrs**	**Prerequisites**
1440241	Ordinary Differential Equations I	3	1440132
1460223	General Geophysics	3	1460120, 1430117
1460231	Introduction to Remote Sensing	3	1430117
1460224	Regional Geology	3	1460220, 1460221
1460xxx	Specialized Elective -1	3	

Year 3, Semester 5 (15 Credits)			
Course	**Title**	**CrHrs**	**Prerequisites**
1460320	Geologic Remote Sensing	3	1460231
1460310	Petrophysics	3	1460220
1460312	Exploration Geophysics I	3	1460223
1460xxx	Specialized Elective -2	3	
1460xxx	Specialized Elective -3	3	

Year 3, Semester 6 (16 Credits)			
Course	**Title**	**CrHrs**	**Prerequisites**
1460313	Exploration Geophysics II	3	1460223
1460311	Petroleum Geology	3	1460220, 1460221
1460xxx	Specialized Elective -4	3	
1460xxx	Specialized Elective -5	3	
0204102	UAE society	3	
1460400	Training	1	146311 & 65 credit hours completed

Year 4, Semester 7 (16 Credits)			
Course	**Title**	**CrHrs**	**Prerequisites**
1460322	Seismology and Plate Tectonic	3	1460223, 1440241
1460420	Well Logging	3	1460310
1460401	Seminar	1	Senior Standing
	University Elective -2	3	
	University Elective -3	3	
	General Free Elective -1	3	

Year 4, Semester 8 (12 Credits)			
Course	**Title**	**CrHrs**	**Prerequisites**
	University Elective -3	3	None
1460402	Graduation Project	3	Department consent
1460413	Reservoir Characterization	3	1460420
	General Free Elective -2	3	

Core courses

Description of the core courses are given below.

1460120	**Physical Geology**	**(2-3:3)**
Introduction to the fundamentals of physical geology. Composition and structure of the Earth, mineral and rock identification, plate tectonics, mountain building, geological structures, earthquakes, volcanism, erosion and sedimentation processes. Laboratory exercises concentrate on mineral and rock identification and the interpretation of topographic and geologic maps. At least one field trip to a nearby locality is required.		

1460220	**Sedimentary Rocks and Sedimentology**	**(3-0:3)**
Elements of sedimentary basin formation, style of sedimentation, provenance, associated facies, and subsequent physicochemical changes through time. Plate tectonic, climatic, allo- and auto-cyclic constraints on sedimentary rocks. Emphasis on convergent and rifted margin sedimentary record. Usage of several macroscopically and microanalytically tools for detailed sedimentary basin analysis. . Computer software will be introduced for basin analysis and data interpretation. At least one field trip is required.		

1460221	**Stratigraphy and Structural Geology**	**(2-3:3)**
This course covers the principal laws of stratigraphy, facies, sea level changes, transgression, regression, sedimentation cyclic, correlation, mass extinction, sequence stratigraphy, surface and subsurface stratigraphy of the UAE. Principles of structural geology. Concepts of true and apparent dip of strata, folds, structural contours for homoclinal and complex surfaces, geological cross-sections, block diagrams, isopachs, faults, intrusive and extrusive igneous structures, impact structures, landslides and sinkholes. Laboratory exercises focus on the interpretation of geological maps and cross-sections and stereographic projection using Schmidt net. Computer software will be used in directional data interpretation, manipulation, and diagram and graph construction. At least one field trip to a nearby locality is required.		

1460223	General Geophysics	(2-3:3)
Introduction to applied and solid-earth geophysics; the gravitational, seismic, magnetic, thermal, and radioactive properties of rocks and earth materials; methods of measurement and their applications to the exploration of the Earth's interior. Physical properties of the earth's interior. Some field trips are required.		

1460224	Regional Geology	(3-0:3)
The course covers the major tectonic elements of the Arabian Peninsula, as well as the rocks and the sedimentary cover in Arabia peninsula. Geological, structural and geomorphological evolution of Arabia with emphasis on hydrocarbon potentials, mineral wealth and underground water resources. At least one field trip is required.		

1460230	Introduction to Geospatial Information Systems (GISs)	(2-3:3)
In this introductory course, students become familiar with the concepts and gain the experience necessary to appreciate the utility of Geographic Information Systems in decision making. Topics covered include the fundamentals of cartography, scale, projection, and coordinate systems in GIS, geo-referencing, data structures, querying, data classification, and basic spatial data analysis. The course provides an overview of the capabilities of GIS software and applications of GIS. Class time is divided between lectures and GIS exercises that reinforce critical concepts.		

1460231	Introduction to Remote Sensing	(2-3:3)
This course describes the fundamental theory and concept of remote sensing. The course would overview properties and capturing techniques of remote sensing data that contain aerial photography, Landsat, Spot, radar imagery and thermal imagery. Basic principles include the basic characteristics of electromagnetic radiation; radiometry; the interactions between radiation and terrestrial materials and atmospheric constituents; characteristics of sensor systems and their measurements. The interactions between radiation and terrestrial materials (Vegetation, soil, water, and rocks) and atmospheric constituents, with incorporation of ancillary data and ground truth.		

1460310	Petrophysics	(2-3:3)
The different types of coring are covered in this course in addition to the handling and storage procedures of core samples. The course covers the methods of measuring the petrophysical characteristics of reservoir rocks such as porosity, permeability, density, resistivity…etc.		

1460311	Petroleum Geology	(2-3:3)
The course includes the definition and properties of petroleum and natural gas. The origin, migration and accumulation of hydrocarbons as related to source, reservoir and seal rocks and reservoir properties. Structural, stratigraphic and combination traps. Different exploration methods and the basin analysis are also introduced as a tool of understanding the course. Computer software will be introduced for basin analysis and data interpretation. At least one field trip is required to investigate the outcrop section of a major reservoir in Saudi Arabia.		

1460312	Exploration Geophysics I	(2-3:3)
The course covers the principles of the seismic reflection and refraction methods with special emphasis on the reflection techniques. It consists of two parts: an introduction to the data acquisition and processing; and the seismic interpretation. The course aims at delineating subsurface Geology including layer succession, types of structures, etc. Also, the course covers an introduction to the attributes as a tool, AVO, VSP and synthetic seismogram. Comparison with well logging data is included. In addition, the basis of the different stacking techniques is introduced.		

1460313	Exploration Geophysics II	(2-3:3)
The course covers the theory and data acquisition and processing of the non-seismic methods, which includes gravity, magnetic, electrical and electromagnetic method. The role of these methods in the exploration for hydrocarbon, groundwater, minerals, as well as civil and environmental applications. Survey execution, interpretation and data processing.		

1460320	Geologic Remote Sensing	(2-3:3)
This course is an introduction to the theory and techniques of remote sensing tools with emphasis on the geosciences and geology. This course explores the use of remote sensing for integrated image interpretation and geological mapping to obtain lithological, stratigraphy and geological structure. It includes image processing techniques, interpretation and analysis that used for geological investigation. The integrated geological mapping approach re-interpreted in a GIS environment on the basis of aerial photographs, satellite imagery and airborne geophysical data. The laboratory work includes the data fusion and processing of satellite images with various spectral and geometric resolutions.		

1460322	Seismology and Plate Tectonics	(2-3:3)
The course covers the dynamics of the solid Earth from theoretical, observational seismology and seismotectonics in relation to earthquake hazard and mitigation. It provides an in-depth study of earthquake seismology and earthquake hazard in relationship with the plate tectonics. The course includes understanding the plates' movements and the procedures of interpreting earthquake seismograms and determination of earthquake focal mechanisms. Examples of the UAE seismicity and the tectonic evolution of the Arabian Plate are given.		

1460400	Training	(1-0:1)
A total period of eight weeks of internship in the industry to gain practical experience in the field of geology. The student is required to submit a written report and make an oral presentation at the department based on the experience of the training program.		

1460401	Seminar	(1-0:1)
Preparation and presentation of selected topics. Each student is expected to submit a written report on his topic and make an oral presentation at the class.		

1460402	Graduation Project	(1-6:3)
Topics will depend on student's and instructor's interest. They may vary from acquisition and interpretation of geophysical data from the field or the laboratory to computer models and simulation of theoretical problems of interest in geophysics, or a mixture of both. Weekly consultations with the instructor as well as a written report are required.		

1460413	Reservoir Characterization	(3-0:3)
Basic petrophysical properties of reservoir rocks including porosity, permeability, fluid saturation, electrical conductivity, capillary pressure, and relative permeability. Laboratory measurement of the reservoir rock characteristics mentioned above.		

1460420	Well Logging	(2-3:3)
Comprehensive study of modem well logging methods, open hole and cased hole log interpretation methods. Production logging. Design of logging programs and examples of applications.		

1460222	Paleontology	(3-0:3)
This course covers the invertebrate fossils groups and their applications in stratigraphy and paleoecology. A special attention is given to their importance in the petroleum exploration in the United Arab Emirates. The course also includes the study of the fossils' characteristic morphology and microstructure and their role in the history of life on Earth.		

1460225	Igneous and Metamorphic Petrology	(3-0:3)
Nature, origin, differentiation and crystallization of magma. Phase relations in silicate melts. Mode of occurrence, textures, petrography and minerals of igneous rocks.		

1460300	Special Topics I	(3-0:3)
The course will cover a special topic in one of the fields of the petroleum, Geology, Remote Sensing and related disciplines. Topics will be selected according to the faculty expertise and the students' interest and enrollment.		

1460301	Special Topics II	(3-0:3)
The course will cover a special topic in one of the fields of the petroleum, Geology, Remote Sensing and related disciplines. Topics will be selected according to the faculty expertise and the students' interest and enrollment.		

1460314	Petroleum Geology of the Middle East	(3-0:3)
This course gives an introductory overview of the Geographic and geomorphologic setting as well as the geologic setting, and the sequence stratigraphy. The History of hydrocarbon exploration is also given as well as the current status of Middle East oil.		

1460324	Hydrogeology	(2-3:3)
In this course, students will become familiar with the concepts of occurrence and movement of groundwater and will gain the knowledge on applications of ground water theories for practical use. Topics covered include the fundamentals of hydrological cycle, types and properties of geological formations holding ground water, theories of ground water movement, usage of flow nets, well hydraulics, analysis of pumping data, groundwater quality and pollution, field exploration and survey for ground water, and usage of computer software		

1460325	Environmental Geology and Risk Management	(3-0:3)
The course covers the environmental problems, hazards and their mitigation. Critical evaluation of geological processes: volcanic activity, earthquakes, slope failures and landslides, flooding, groundwater movement, solution cavities and sinkholes. Environmental problems associated with human interaction: groundwater pollution, groundwater withdrawal, acid rain, solid waste disposal, land development and urbanization, agricultural activity, soil erosion, and desertification. Current environmental issues. Selected case studies and computer software related to the subject will be introduced At least one field trip to a nearby locality is required.		

1460414	Petroleum Geochemistry	(3-0:3)
The course's main goal is to provide the student with an overview of the petroleum industry: its history, its technical achievements, its role in the global-economy and its future prospects. Introduction to modern exploration, production and processing operations is included as well as highlights of the petrochemicals and petroleum industry.		

1460415	Basin Analysis	(3-0:3)
The mystery of hydrocarbon accumulation in sedimentary basins is discussed with reference to worldwide and Arabian Gulf examples. The course includes a special emphasis on the key elements and the where about and how to find the accumulation.		

1460421	Seismic Stratigraphy	(3-0:3)
This course covers the procedure of understanding the stratigraphy by using the seismic sections. It covers the key elements of seismic reflection patterns, amplitudes and frequencies essential to a specific basin stratigraphic and facies analysis that led to oil and gas play definition. Also, the geological description and interpretation of seismic parameters within the seismic- sequence correlation framework are covered. The method of analyzing depositional environment is being detailed with emphasis on carbonate environments, type of rocks, Sedimentary sequences, fluid content and contacts, palaeo-sea level change and the Geologic history.		

1460426	Advanced Geographic Information Systems	(2-3:3)
The course deals with design and operation and operation of Geospatial Information Systems (GISs) and their role in digital mapping and spatial data management. Topics covered include basic data structure, data source and models, geospatial analysis, digital elevation data and terrain analysis. The Course includes an embedded computer lab for hands-on training with GIS software. It includes the use of GIS within oil and gas exploration and production activities using petroleum industry spatial data and workflows. It Introduce the functionality that allows geoscientists to import spatial and non-spatial databases, integrate, manage and analyze data to produce information for decision-making.		

1460431	Environmental Remote Sensing	(2-3:3)
This course is designed to introduce students to remote sensing science and technology. It emphasizes mastering fundamental remote sensing concepts and utilizing remotely sensed data for environmental information extraction and problem solving. The course introduces the student to remote sensing basics for environmental Modeling and monitoring and its application in geoscience. The first part of the course will cover the remote sensing and techniques to acquire, enhance, interpret and analyze remote sensing imagery using visual and computer-based methods. The second part of the course addresses the role and nature of Environmental Models. The third part deals with the application of remote sensing principles and data to environmental science. Topics include the use of remote sensing for environmental applications related to different studies of hazards monitoring, hydrology, air pollution, and land use/land cover.		

1460432	Digital Image Processing	(3-0:3)
This course explores the major categories of digital image processing. Digital image fundamentals, perception, discrimination, and sampling. Image transforms using FFT Fourier Algorithm. Image rectification and restoration, image enhancement, filtering and multi-image manipulation, image encoding, image segmentation and description. Image classifications, typical steps in numerical analysis. Classification accuracy assessment and data merging training on a selected software such as Mat lab.		